

Course Code: CS1385		
Name of the Programme: BCA (Hons.)		
Name of the Course: Web and UX Design		
Semester: I		
Course Credit	No. of Hours Per Week (L + T + P)	Total No. of Teaching Hours
3	1:1:2	60
Course Objectives:		
a) Introduce the core technologies and principles of modern web development.		
b) Develop the ability to design and create user-centered, accessible websites.		
c) Familiarize students with the UX design process, including planning, prototyping, and usability testing.		
d) Provide hands-on experience in implementing and testing web interfaces, laying a foundation for further learning and professional growth.		

Course outline (Syllabus of the course)

Module - 1: Web and HTML Fundamentals	No. of Hours
	3 Hrs.
Introduction to the Web and Web Development, Basic Structure of an HTML Document, Basic HTML Tag and Attributes, Creating Links, Lists, Tables, and Forms	
Module - 2 : CSS Fundamentals	No. of Hours
	3 Hrs.
CSS Selectors, Styling Text, Boxes, and Layouts, Introduction to Responsive Design.	
Module - 3 : JavaScript Basics	No. of Hours
	3 Hrs.
JavaScript Syntax and Programming Concepts, Manipulating the DOM (Document Object Model).	
Module - 4 : UX Design Fundamentals	No. of Hours
	3 Hrs.
Introduction to UX Design and Its Importance, User Research and Personas, Wireframing and Prototyping, Usability Testing and Basic Accessibility Standards.	
Module - 5 : Integrated Project	No. of Hours
	3 Hrs.
Overview of Project Requirements and Planning, Integration of HTML, CSS, JavaScript, and UX Design Principles and Project Management.	

Course Outcomes:

CO1 Understand the basic structure of HTML and apply CSS to design well-structured and Visually styled web pages.

CO2 Analyze and manipulate the Document Object Model (DOM) using JavaScript to create dynamic and interactive web pages.

CO3 Understand the principles of user experience (UX) design to create user-friendly and accessible web interfaces.

CO4 Apply web development and UX principles to develop a complete web project from concept to implementation.

Text Books:

1	Duckett, J. (2011). <i>HTML and CSS: Design and build websites</i> (1st ed.). John Wiley & Sons. ISBN: 978-1-118-00818-8
2	Duckett, J. (2014). <i>JavaScript and jQuery: Interactive front-end web development</i> . John Wiley & Sons. ISBN: 978-1-118-53164-8 / 978-1-118-87165-2
3	Unger, R., & Chandler, C. (2012). <i>A project guide to UX design: For user experience designers in the field or in the making</i> (2nd ed.). New Riders. ISBN: 978-0-321-81538-5

Reference Books:

1	Robbins, J. N. (2018). <i>Learning web design: A beginner's guide to HTML, CSS, JavaScript, and web graphics</i> . Shroff/O'Reilly. ISBN: 978-1-491-46020-2
2	Haverbeke, M. (2011). <i>Eloquent JavaScript: A modern introduction to programming</i> . No Starch Press. ISBN: 978-1-59327-282-1
3	Norman, D. A. (2013). <i>The design of everyday things</i> (2nd ed.). Basic Books. ISBN: 978-0-465-05065-9
4	Gothelf, J., & Seiden, J. (2016). <i>Lean UX: Designing great products with agile teams</i> . O'Reilly Media, Inc. ISBN: 978-1-491-95360-0
5	Krug, S. (2014). <i>Don't make me think, revisited: A common sense approach to web usability</i> . New Riders. ISBN: 978-0-321-96551-6

Tutorial Components	
1	Introduction to Web Development: Learn how the web works, the roles of browsers and servers, and create your first HTML webpage.
2	HTML Structure and Basic Tags: Understand HTML elements, attributes, and semantic tags to structure webpages effectively.
3	Organizing Content: Use tables, lists, and forms to organize and collect data on web pages.
4	Introduction to CSS: Explore CSS syntax, selectors, and ways to style webpages using inline, internal, and external methods.
5	Styling Text and Colors: Apply fonts, text styles, colors, and background designs to enhance webpage aesthetics.
6	CSS Layout Principles: Master the box model, positioning, and flexbox to create structured webpage layouts.
7	Responsive Design: Use media queries and fluid layouts to make webpages mobile-friendly.
8	JavaScript Syntax Basics: Learn variables, operators, and control structures to write simple JavaScript code.
9	DOM Manipulation: Modify HTML content dynamically using the Document Object Model.
10	Event Handling: Add interactivity with event listeners like clicks, keypresses, and form validations.
11	Introduction to UX Design: Understand user-centered design and the importance of creating user-friendly interfaces.
12	User Research: Conduct surveys and interviews to gather insights for designing better webpages.
13	Wireframing and Prototyping: Plan layouts and navigation using tools to create low-fidelity wireframes.
14	Accessibility in Web Design: Follow WCAG standards to make webpages inclusive for all users.
15	Integrated Project: Combine HTML, CSS, JavaScript, and UX principles to build a complete, functional, and visually appealing webpage.

Laboratory Components

1	Write a program to develop Basic Webpage with HTML.
2	Write a program to develop creating a personal profile page.
3	Write a program to design a feedback form using various HTML form elements.
4	Write a program to build a table to display a schedule.
5	Write a program to style a basic webpage with different font styles, colors, and background images.
6	Write a program to create a two-column layout using floats and a grid system.
7	Write a program to make a web page responsive using media queries.
8	Write a program to style a form and table for better usability and appearance.
9	Write a JavaScript program to perform basic arithmetic operations.
10	Write a JavaScript program to Manipulate the content of HTML elements.
11	Write a program to create an interactive form that validates user input.
12	Build a simple to-do list application with dynamic DOM manipulation
13	Write a program to create simple interactive web pages using HTML, CSS, JavaScript.
14	Write a program to design a sitemap and navigation structure using HTML, CSS.
15	Write a program to create a basic webpage with a heading, paragraph, and image.
16	Write a program to create a profile page with biography, contact information, and social media links using HTML and CSS. Note: All pages should have consistent styling.
17	Write a program to design a feedback form using various form elements with HTML and CSS.

Name of the Programme:	BSc(H)	
Title of the Course:	Web and UX Design	
Course code	CS1385	
Semester:	I	
Names of the Course Instructor(s)	Dr. V Balaji Prof. Mohammed Danish	
Course credit	No. of hours per week (L + T + P) (1:1:2)	Total no. of Teaching hours
3	4	60

Session	Module No.	Topic	Pedagogy /activities	Pre-reading/reference
1	1	Introduction to Web Development.	Explanation using browser-server diagrams, live demo of webpage rendering.	Duckett (2011), Ch. 1
2		HTML Structure and Basic Tags.	Code walkthrough of HTML5 document, tag usage.	Duckett (2011), Ch. 2
3		Organizing Content: Tables, Lists, Forms.	Hands-on practice and demo.	Duckett (2011), Ch. 6, Ch. 7, Ch. 9, Ch. 12
4		Tutorial 1: Introduction to Web Development: Learn how the web works, the roles of browsers and servers, and create your first HTML webpage.	Given a template of Web Page, students are instructed to develop a similar one.	External Resources.

5		Tutorial 2:HTML Structure and Basic Tags: Understand HTML elements, attributes, and semantic tags to structure web pages effectively.	Given a template of Web Page, students are instructed to develop a similar one.	External Resources.
6		Tutorial 3:Organizing Content: Use tables, lists, and forms to organize and collect data on web pages.	Given a template of Web Page, students are instructed to develop a similar one.	External Resources.
7		Lab 1: Write a program to develop Basic Webpage with HTML.	Create a basic webpage structure using HTML tags including headings, paragraphs, and images.	Duckett (2011), Ch. 1, 2, 6, 7, 9, 12
8				
9		Lab 2: Write a program to develop creating a personal profile page.	Design a personal profile page with sections like bio, photo, and contact details using semantic HTML.	Duckett (2011), Ch. 1, 2, 6, 7, 9, 12
10				
11		Lab 3: Write a program to design a feedback form using various HTML form elements.	Building a feedback form with text fields, radio buttons, checkboxes, and submit/reset functionality.	Duckett (2011), Ch. 1, 2, 6, 7, 9, 12
12				
13		Lab 4: Write a program to build a table to display a schedule.	Demonstrate how to create an HTML table to display a weekly	Duckett (2011), Ch. 1, 2, 6, 7, 9, 12

14			schedule using proper row and column tags.	
15	2	Introduction to CSS, Styling Text and Colors	Demo of selectors and declarations. Apply fonts, backgrounds for the web page.	Duckett (2011), Ch. 10,11,13
16		CSS Layout Principles	Box Model Illustrations.	Duckett (2011), Ch. 15,16
17		Responsive Design	Media queries, device simulation.	Duckett (2011), Ch. 17
18		Tutorial 1: Introduction to CSS: Explore CSS syntax, selectors, and ways to style webpages using inline, internal, and external methods.	Given a template of Web Page, students are instructed to develop a similar one.	External Resources.
19		Tutorial 2: Styling Text and Colors: Apply fonts, text styles, colors, and background designs to enhance webpage aesthetics.	Given a template of Web Page, students are instructed to develop a similar one.	External Resources.
20		Tutorial 3: CSS Layout Principles: Master the box model, positioning, and flexbox to create structured webpage layouts.	Given a template of Web Page, students are instructed to develop a similar one.	External Resources.
21		Tutorial 4: Responsive Design: Use media queries and fluid layouts to make webpages mobile friendly.	Given a template of Web Page, students are instructed to develop a similar one.	External Resources.

22		Lab 5: Write a program to style a basic webpage with different font styles, colors, and background images.	Applying custom font families, text styles, color schemes, and background images using CSS to enhance the visual appeal of a basic HTML page.	Duckett (2011), Ch. 11, 13, 15, 16, 17
23				
24		Lab 6: Write a program to create a two-column layout using floats and a grid system.	Demonstrate how to implement a two-column layout using CSS float and grid properties, followed by hands-on practice to build a structured content layout.	Duckett (2011), Ch. 11, 13, 15, 16, 17
25				
26		Lab 7: Write a program to make a web page responsive using media queries.	Apply media queries to adjust layout and typography across different screen sizes, enabling them to build a mobile-responsive webpage.	Duckett (2011), Ch. 11, 13, 15, 16, 17
27				
28		Lab 8: Write a program to style a form and table for better usability and appearance.	To enhance the appearance and usability of HTML forms and tables through CSS styling techniques such as padding, borders, hover	Duckett (2011), Ch. 11, 13, 15, 16, 17
29				

			effects, and alignment.	
30	3	JavaScript Syntax Basics.	Coding and debug walkthrough.	Duckett (2014), Ch. 2
31		DOM Manipulation.	Code-along DOM access and update.	Duckett (2014), Ch. 5
32		Event Handling.	Code to add Button clicks, keyboard events.	Duckett (2014), Ch. 6
33		Tutorial 1:JavaScript Syntax Basics: Learn variables, operators, and control structures to write simple JavaScript code.	Given a template of Web Page, students are instructed to develop a similar one.	External Resources.
34		Tutorial 2:DOM Manipulation: Modify HTML content dynamically using the Document Object Model.	Given a template of Web Page, students are instructed to develop a similar one.	External Resources.
35		Tutorial 3:Event Handling: Add interactivity with event listeners like clicks, keypresses, and form validations.	Given a template of Web Page, students are instructed to develop a similar one.	External Resources.
36		Lab 9:Write a JavaScript program to perform basic arithmetic operations.	Guide students to write a JavaScript program that takes user input and performs basic arithmetic operations like addition, subtraction, multiplication,	Duckett (2014), Ch. 2, 4, 5, 6
37				

			and division.	
38		Lab 10: Write a JavaScript program to Manipulate the content of HTML elements.	Demonstrate DOM manipulation by updating text, styles, or attributes of HTML elements using JavaScript event handling.	Duckett (2014), Ch. 2, 4, 5, 6
39				
40			Create a form with real-time validation using JavaScript to check input types, required fields, and display error messages.	Duckett (2014), Ch. 2, 4, 5, 6
41				
42			Facilitate the development of a dynamic to-do list where users can add, remove, and mark tasks using JavaScript and DOM methods.	Duckett (2014), Ch. 2, 4, 5, 6
43				
44	4	Introduction to UX Design.	Whiteboard+Infographics Discussion on examples of Bad vs good UX.	Unger & Chandler (2012), Ch. 1
45		User Research.	Whiteboard, Infographics.	Unger & Chandler (2012), Ch. 3

46		Wireframing Basics.	Whiteboard, Infographics.	Unger & Chandler (2012), Ch. 6
47		Prototyping Tools, Accessibility Guidelines.	Use online tools to convert wireframes. Whiteboard, Infographics.	Unger & Chandler (2012), Ch. 7, Robbins (2018), Ch. 14
48		Tutorial 1: Introduction to UX Design: Understand user-centered design and the importance of creating user-friendly interfaces.	Given a template of Web Page, students are instructed to develop a similar one.	External Resources.
49		Tutorial 2: User Research: Conduct surveys and interviews to gather insights for designing better webpages.	Given a template of Web Page, students are instructed to develop a similar one.	External Resources.
50		Tutorial 3: Wireframing and Prototyping: Plan layouts and navigation using tools to create low fidelity wireframes.	Given a template of Web Page, students are instructed to develop a similar one.	External Resources.
51		Lab 13: Write a program to design a sitemap and navigation structure using HTML, CSS.	To design a clear sitemap and responsive navigation menu using semantic HTML and styled with CSS for usability.	Unger & Chandler (2012), Ch. 1, 3, 6, 7; Robbins (2018), Ch. 14
52				
53		Lab 14: Write a program to create a profile page with biography, contact information, and social media links using	To create a well-structured profile page featuring biography, contact details,	Unger & Chandler (2012), Ch. 1, 3, 6, 7; Robbins (2018), Ch. 14
54				

		HTML and CSS.	and styled social media links using HTML and CSS.	
55		Lab 15: Write a program to create a profile page with biography, contact information, and social media links using HTML and CSS.	Develop a polished and visually appealing personal profile page integrating biography, contact information, and social media icons with CSS design principles.	Unger & Chandler (2012), Ch. 1, 3, 6, 7; Robbins (2018), Ch. 14
56				
57	5	Integrated Project	To develop a complete and functional front-end website.	External Resources
58				
59				
60				

Assessment and Evaluation

Continuous Internal Evaluation (CIE) :70%	
Components of CIE	Weightage of each component
CIE-1	
- Quiz Graded Component 1	10%
- Mini Project Phase -1	10 %
CIE-2	
- Written Test	25 %
CIE-3	
- Viva	5 %
- Day-to-day Execution	10 %
- Mini Project Phase - 2	10 %

Semester End Exams SEE: 30 Marks	
Mode of Exam	Practical
Weightage of the exam	30 %

Course Outcomes- Programme Outcomes Matrix

Sl. No.	Course Outcome	PO1	PO2	PO3	PO4	PO5
1	Understand the fundamentals of HTML, CSS, and JavaScript for web development	3	2	3	-	-
2	Design and develop responsive, user-friendly web pages and interfaces	3	3	3	2	2
3	Apply JavaScript to create dynamic and interactive user experiences	2	3	3	1	-
4	Implement UX design principles, including wireframing, personas, and usability testing	2	3	2	3	2
5	Develop and present a complete web design project demonstrating real-world usability and accessibility	2	3	3	3	3

Rubrics for evaluation of CIE (separate for each CIE component)

Total Marks		20
CIE-1 Components		Rubrics for Assessment
CIE-1	One mark Questions Graded Component -1	One mark questions will be conducted for 10 marks. Each right answer carries one mark each. No negative marking for wrong answers. (10M)
	Mini Project Phase 1	Design and develop a responsive, user-friendly, and visually appealing website with a focus on both User Interface (UI) and User Experience (UX) principles. (Refer to Table-1) (10M)

Table-1:

Graded Component-2 (10 Marks):

Criteria	Excellent	Satisfactory	Poor
Problem Statement Understanding. (2M)	Clear, well-defined objective. Content aligns perfectly with the problem statement. (2M)	Objective is mostly clear. Some minor mismatches in content or scope. (1M)	Problem statement misunderstood or not reflected in the webpage content. (0M)
HTML Structure & Semantics. (3M)	Proper use of all relevant HTML tags (e.g., <header>, <nav>, <main>, <footer>, etc.). (2M)	Mostly correct use of HTML tags. A few missing or misplaced tags. (1M)	Poor or incorrect use of HTML tags. Structure is unclear or invalid. (0M)
Layout and Content Organization. (2M)	Logical, clean layout. Content sections are well-separated and easy to follow. (2M)	Layout is understandable but may be slightly cluttered or inconsistent. (1M)	Disorganized layout. Difficult to follow structure or locate content. (0M)
Navigation (if applicable). (2M)	Clear, working navigation links or menus. Easy to move between sections. (2M)	Basic navigation present but may have issues or limited functionality. (1M)	No navigation, or broken/missing links. (0M)
Visual Appeal (Basic UI). (2M)	Good use of spacing, alignment, and basic visual hierarchy. Simple but effective. (2M)	Visually acceptable but lacks consistency or polish. (1M)	Unattractive layout, poor alignment, inconsistent fonts or element placement. (0M)

Total Marks		25
CIE-2 Components		Rubrics for Assessment
CIE-2	Theory	Test will be conducted for 25 marks and will be evaluated according to the Scheme of Evaluation prepared by the team of course faculty.

Total Marks		25
CIE-3 Components		Rubrics for Assessment
CIE-3	Practical Component	Day to Day Execution = 10 marks (<i>Refer Table-2</i>) Viva = 5 marks (<i>Refer Table-2</i>) Mini Project Phase-2=10 marks (<i>Refer Table-4</i>)

Table-2

Lab Program Execution rubrics (Max: 6 marks)					
Sl. No	Criteria	Measuring Methods	Excellent	Good	Poor
1	Layout / Design (2M)	Observations	The web site has an Exceptionally attractive and usable layout. It is easy to locate all important elements. White space, graphic elements and/or alignment are used effectively to organize material. (2 M)	The web site has a Usable layout, but may appear busy or boring. It is easy to locate most of the important elements. (<2 M and >=1 M)	The web site has a Cluttered look and is Confusing. It is often difficult to locate Important elements. (0 M)

2	Results and Documentation (2M)	Observations	The HTML code is fully HTML 5.0 compliant. Each web page has a title defined in the head section. All images are visible and have size < 25K on the disk. (2 M)	The HTML 5.0 has few minor deficiencies. (1 M)	The HTML 5.0 code has multiple major Deficiencies. (0 M)
3	Deadline (2M)	Timestamp of submission, file check	Submitted on or before deadline. (2M)	Submitted slightly late (<24 hours). (1M)	Submitted >24 hours late or incomplete submission. (0M)
Viva Voce Rubrics (Max: 4 marks)					
1	Conceptual Understanding (2M)	Viva Voce	Explains thoroughly the concepts. (2 M)	Adequately explains the concepts. (1 M)	Unable to explain the concept. (0 M)
2	Use of appropriate Design Techniques (1M)	Viva Voce	Insightful explanation of appropriate design techniques. (1 M)	Sufficiently explains the use of appropriate design techniques. (0.5 M)	Unable to explain the design techniques. (0 M)
3	Communication of Concepts (1M)	Viva Voce	Communicates the concept used in problem solving well. (1 M)	Sufficiently communicates the concepts used in problem solving. (0.5 M)	Unable to communicate the concepts used in the concepts. (0 M)

Table-3

Rubrics for Mini Project Phase 2 (Total: 10 Marks)					
Sl. No	Criteria	Excellent	Very Good	Good	Poor
1	Functionality & Completion. (4M)	Fully functional, all features implemented, no bugs, meets all requirements. (4M)	Mostly complete, minor missing features or small bugs. (3M)	Basic functionality; few features missing or not working. (2M)	Incomplete or mostly non-functional. (0M)

2	Design & Responsiveness. (2M)	Clean, responsive UI; consistent styling across pages. (2M)	Mostly good design; minor issues with responsiveness or layout. (1M)	Basic design, lacks responsiveness or consistency. (1M)	Poor or no styling, non-responsive. (0M)
3	Code Quality (2M)	Clean, modular, semantic HTML/CSS/JS; follows best practices. (2M)	Organized code with some duplication or style issues. (1M)	Basic structure, some poor coding practices. (1M)	Poorly written, unstructured code. (0M)
4	Presentation & Viva (2M)	Confident explanation of design & logic; good handling of questions. (2M)	Clear explanation with minor gaps. (1M)	Basic explanation; lacks depth. (1M)	Unable to explain project well. (0M)

Table-4

Grading Rubrics for Web UX & Design SEE (Total: 30 Marks)					
Sl. No	Criteria	Excellent	Very Good	Good	Poor
1.	Implementation (10 M)	Complete end-to-end responsive web design with innovative UI/UX elements, accessibility, and performance optimization. (9-10 M)	Fully functional website with all key design and development features implemented (visual hierarchy, navigation, responsive design). (7-8 M)	Basic website with limited design elements or minor functional gaps. (5-6 M)	Incomplete, non-functional, or poorly designed website. (0-4 M)
2.	Design & Code Quality (5 M)	Clean, modular, semantic HTML/CSS/JS with clear	Well-organized and readable code with good use of design	Basic code with some design flaws, limited styling, or inconsistent user	Poorly written, disorganized, or non-functional code; major

		structure, accessibility, and consistency; advanced design principles applied. (5 M)	patterns and minor inconsistencies. (4-3 M)	interface. (2-1 M)	design issues. (0 M)
3.	User Experience (UX) Report(5 M)	Comprehensive, well-structured report detailing design choices, user journey, wireframes/mock ups, and usability considerations. (5 M)	Clear and complete report with thoughtful analysis; minor gaps in detailing or design rationale. (4-3 M)	Report covers essentials (design steps, basic reasoning) but lacks depth or proper structure. (2-1 M)	Incomplete, unclear, or superficial report with limited understanding of UX concepts. (0 M)
4.	Viva (Presentation & Explanation) (10 M)	Confident and articulate explanation of design decisions, user flow, and technical choices; effectively answers design and implementation questions. (9-10 M)	Good presentation with clear understanding; minor gaps in explanation or responses. (7-8 M)	Basic understanding; some difficulty in explaining design choices or user flow. (5-6 M)	Weak explanation, poor understanding of design principles, or inability to communicate project decisions. (0-4 M)

Student list (template – 3) (extracted from SAP)

Section - A

Sl. No.	USN No.	Student name

Section - B

Sl. No.	USN No.	Student name

Signature of the Course Faculty

- 1.
- 2.